Answer keys at the end, Some tricky Q explained at the end
MCQ set **(Berkovitz)** —

Alveolar Bone: Academic Assessment (Part I)

1. **The alveolar process is a tooth-dependent structure of the maxilla and mandible. Following the extraction of a tooth, what is the primary physiological response of this bone?**
   A. Increased mineralization density
   B. Maintenance of structural volume
   C. Accelerated vascular proliferation
   D. Progressive resorption and atrophy
   E. Conversion into dense cortical bone only
2. **The "alveolar bone proper" constitutes the internal lining of the tooth socket. Which clinical term is used to describe this structure on a radiograph?**
   A. Trabecular bone
   B. Cortical plate
   C. Spongy bone
   D. Bundle bone
   E. Lamina dura
3. **In terms of chemical composition, what is the approximate percentage by weight of the inorganic (mineralized) phase in mature bone?**
   A. 45%
   B. 50%
   C. 60%
   D. 67%
   E. 70%
4. **Osteoblasts are specialized cells derived from mesenchymal origins. Which histological feature is characteristic of an active, secreting osteoblast?**
   A. High lysosomal concentration
   B. Abundant rough endoplasmic reticulum and a prominent Golgi complex
   C. Multicellular syncytium appearance
   D. Localization within a Howship’s lacuna
   E. Origin from the hematopoietic monocyte line
5. **The alveolar bone proper is frequently referred to as "bundle bone." This nomenclature is specifically due to:**
   A. The clustering of trabecular bone
   B. The presence of concentric lamellae
   C. High levels of intrinsic vascularity
   D. The embedding of the principal fibers of the periodontal ligament (Sharpey’s fibers)
   E. The presence of cementum bundles
6. **The alveolar bone proper is also termed the "cribriform plate." This designation refers to which structural characteristic?**
   A. A perfectly smooth internal surface
   B. The extreme degree of mineralization
   C. The developmental origin from the dental follicle
   D. The presence of multiple Volkmann’s canals for neurovascular passage
   E. The lack of intervening marrow spaces
7. **Osteocytes are mature bone cells situated within lacunae. Through which structural mechanism do these cells maintain intercellular communication?**

A. Direct diffusion through the marrow
B. Through the periodontal ligament space
C. Via cytoplasmic processes located within canaliculi
D. Through active migration within the calcified matrix
E. Via gap junctions on the periosteal surface only

8. **Cancellous bone (supporting bone) is located between the cortical plates and the alveolar bone proper. In which anatomical site is this trabecular bone most significantly present?**
   A. Anterior mandibular region
   B. Maxillary molar region
   C. Maxillary incisor region
   D. Labial aspect of mandibular canines
   E. Midline of the hard palate
9. **Osteoclasts are the primary cells responsible for bone degradation. In which specific micro-environment are these cells typically located during active resorption?**
   A. Howship’s lacunae
   B. Volkmann's canals
   C. Outer surface of the periosteum
   D. Internal marrow spaces only
   E. Along the insertion of Sharpey's fibers
10. **Before mineralization occurs, osteoblasts secrete an organic matrix known as:**
    A. Chondroid
    B. Cementoid
    C. Dentinoid
    *D. Enamel matrix
    E. Osteoid
11. **The cortical plates of the alveolar process exhibit variations in thickness. In which region are these plates generally most substantial?**
    A. Mandibular molar region
    B. Maxillary incisor region
    C. Floor of the orbit
    D. Nasal floor
    E. Sublingual fossa
12. **In a clinically healthy periodontium, where is the alveolar crest typically situated relative to the cemento-enamel junction (CEJ)?**
    A. Directly at the CEJ
    B. 5 mm apical to the CEJ
    C. 1.5 to 2.0 mm apical to the CEJ
    D. Coronal to the CEJ
    E. Level with the proximal contact point
13. **The "Haversian system," or osteon, is the fundamental structural unit of which bone type?**
    A. Compact (Cortical) bone
    B. Cancellous (Spongy) bone
    C. Bundle bone
    D. Woven bone
    E. Trabecular bone
14. **During orthodontic tooth movement, which cell type demonstrates increased activity on the "pressure side" of the periodontal ligament?**
    A. Osteocytes
    B. Fibroblasts
    C. Cementoblasts
    D. Osteoclasts
    E. Ameloblasts

15. **Which molecular component constitutes approximately 90% of the organic matrix of bone?**
    A. Elastin
    B. Type I Collagen
    C. Type IV Collagen
    D. Glycosaminoglycans
    E. Proteoglycans
16. **The term "Dehiscence" refers to a specific defect in the alveolar bone characterized by:**
    A. A circumscribed "window" in the bone with an intact margin
    B. A generalized thinning of the trabeculae
    C. A V-shaped or cleft-like loss of cortical bone including the marginal edge
    D. A localized thickening of the bundle bone
    E. The presence of accessory nutrient canals
17. **Which enzymatic marker is used to identify active osteoblasts during the bone formation phase?**
    A. Acid phosphatase
    B. Tartrate-resistant acid phosphatase (TRAP)
    C. Alkaline phosphatase
    D. Hyaluronidase
    E. Matrix metalloproteinase
18. **"Woven bone" is characterized by a disorganized collagen fiber arrangement. In which physiological scenario is this bone type predominantly formed?**
    A. Mature alveolar bone proper in adults
    B. Secondary remodeling of cortical plates
    C. Midline of the mandible in late adulthood
    D. Rapid wound healing or initial fracture repair
    E. Formation of the secondary palate only
19. **The inorganic mineral phase of the alveolar process is primarily composed of which crystal type?**
    A. Calcium carbonate
    B. Magnesium sulfate
    C. Sodium chloride
    D. Fluorapatite
    E. Hydroxyapatite
20. **A "Fenestration" in the alveolar process is defined as:**
    A. A localized bony exostosis
    B. A circumscribed "window" in the cortical plate exposing the root surface
    C. The natural porosity of the alveolar bone proper
    D. An acute fracture of the alveolar crest
    E. The radiographic appearance of the lamina dura
21. **While the mineral phase provides hardness, the organic collagenous component of bone primarily contributes:**
    A. Radiopacity
    B. Resistance to compressive forces
    C. Tensile strength and resilience
    D. Protection against microbial invasion
    E. Structural pigmentation
22. **Osteocytes are interconnected via gap junctions at the terminal ends of their processes. What is the functional priority of this network?**
    A. To facilitate bone resorption
    B. To anchor the tooth into the socket
    C. To inhibit neural conduction
    D. To allow for the exchange of nutrients and signaling molecules
    E. To synthesize new osteoid matrix

23. **The "Endosteum" is a thin connective tissue membrane that serves as a lining for:**
    A. The external maxillary surface
    B. The periodontal ligament space
    C. The pulp chamber
    D. Internal marrow spaces and vascular canals
    E. The gingival sulcus
24. **When orthodontic tension is applied to a tooth, bone deposition occurs on the tension side due to the recruitment of:**
    A. Osteocytes
    B. Osteoclasts
    C. Osteoblasts
    D. Macrophages
    E. Cementocytes
25. **Sharpey’s fibers within the alveolar bone proper are histologically identified as:**
    A. Elastic fibers of gingival origin
    B. Mineralized continuations of the principal periodontal ligament fibers
    C. Sensory nerve fibers of the pulp
    D. Muscular attachments for the facial muscles
    E. Vascular structures within Volkmann's canals
26. **From which embryonic structure does the alveolar process originate during development?**
    A. Dental papilla
    B. Enamel organ
    C. Dental lamina
    D. Vestibular lamina
    E. Dental follicle (Dental sac)
27. **"Reversal lines" observed in bone histology are indicative of:**
    A. The site where bone resorption ceased and formation commenced
    B. Continuous and uninterrupted bone growth
    C. The precise boundary between bone and cementum
    D. Locations of current osteoclastic activity
    E. Chronological age of the specimen
28. **In the mandibular arch, supporting cancellous bone is frequently absent or highly attenuated in which specific area?**
    A. Labial aspect of the anterior teeth
    B. Lingual aspect of the molar teeth
    C. Retromolar pad region
    D. Mandibular angle
    E. Mental protuberance
29. **What do "resting lines" signify within a histological section of bone?**
    A. Periods of rapid bone destruction
    B. The orientation of neurovascular bundles
    C. The exclusive presence of bundle bone
    D. Rhythmic or cyclic periods of bone deposition
    E. The path of Volkmann’s canals
30. **Bone is considered a dynamic tissue. This characteristic describes its inherent ability to:**
    A. Alter mineral content on a daily cycle
    B. Undergo constant remodeling through coordinated resorption and apposition
    C. Transition into cementum under pathological stress
    D. Move independently of the dental structures
    E. Maintain growth in the absence of a vascular supply

Answer Key (1–30)

Q. A	Q. A	Q. A	Q. A	Q.A
1. E	**2.** E	**3.** E	**4.** B	**5.** D
6. D	**7.** C	**8.** B	**9.** A	**10.** E
11. A	**12.** C	**13.** A	**14.** D	**15.** B
16. C	**17.** C	**18.** D	**19.** E	**20.** B
21. C	**22.** D	**23.** D	**24.** C	**25.** B
26. E	**27.** A	**28.** A	**29.** D	**30.** B

Alveolar Bone: Academic Assessment Part II

(thora sa broad anatomical tarka so do it in the end 😉)

31. **The alveolar process is composed of two main parts: the alveolar bone proper and the supporting alveolar bone. Which of the following describes the supporting bone?**
    A. It is exclusively composed of bundle bone.
    B. It consists of the cortical plates and the spontaneous bone.
    C. It lacks any vascular supply.

D. It consists of the cortical plates and the intervening cancellous (spongy) bone.
E. It is the only part of the jaw that does not resorb after extraction.

32. **The cortical plates of the alveolar process are continuous with which of the following structures at the alveolar crest?**
    A. The periodontal ligament.
    B. The alveolar bone proper.
    C. The cementum of the tooth.
    D. The enamel-dentine junction.
    E. The dental follicle.
33. **In the histology of bone, "interstitial lamellae" represent which of the following?**
    A. New bone currently being formed by osteoblasts.
    B. Unmineralized osteoid.
    C. Small canals for blood vessels.
    D. Remnants of older Haversian systems left behind during remodeling.
    E. The connection between Sharpey’s fibers and the matrix.
34. **Which of the following describes the arrangement of collagen fibers in "lamellar bone"?**
    A. Haphazard and random distribution.
    B. Highly organized into parallel layers or sheets.
    C. Absent in the mineralized matrix.
    D. Only found surrounding the Volkmann's canals.
    E. Vertical to the long axis of the bone.
35. **The "Lamina Dura" observed on a periapical radiograph corresponds to which histological structure?**
    A. The cortical plate of the mandible.
    B. The trabecular bone of the maxilla.
    C. The endosteum.
    D. The alveolar bone proper (Cribriform plate).
    E. The cemento-enamel junction.
36. **Which of the following cell types is derived from the fusion of mononuclear cells of the hematopoietic lineage (monocyte/macrophage)?**
    A. Osteoblast.
    B. Osteocyte.
    C. Fibroblast.
    D. Undifferentiated mesenchymal cell.
    E. Osteoclast.
37. **The process of "Intramembranous Ossification" is the primary mode of development for which of the following?**
    A. The long bones of the limbs.
    B. The vertebrae.
    C. The alveolar process and the bones of the cranial vault.
    D. The head of the condyle only.
    E. The hyoid bone.
38. **What is the primary function of the "Canaliculi" within the mineralized bone matrix?**
    A. To provide a passage for oxygen and nutrients to reach trapped osteocytes.
    B. To house the primary blood vessels of the bone.
    C. To serve as a site for Sharpey's fiber insertion.
    D. To facilitate the movement of osteoclasts.
    E. To store calcium carbonate.
39. **Which of the following best describes "Bundle Bone"?**
    A. That part of the alveolar bone proper where Sharpey’s fibers of the PDL are embedded.
    B. Bone that contains no collagen fibers.

C. Bone found only in the center of the marrow spaces.
D. The thickest portion of the cortical plate.
E. Bone that forms during embryonic life and never remodels.

40. **The "Alveolar Crest" is most accurately described as the junction between:**
A. The cortical plate and the alveolar bone proper.
B. The enamel and the cementum.
C. The sublingual and submandibular fossae.
D. The primary and secondary dentine.
E. The trabeculae and the marrow.
41. **(Tricky) In a case of chronic periodontitis, the "Lamina Dura" may appear broken or absent on a radiograph. Histologically, this indicates destruction of:**
A. The cortical plate of the jaw.
B. The cancellous bone.
C. The cribriform plate (alveolar bone proper).
D. The cementum.
E. The periodontal ligament fibers only.
42. **(Tricky) If a tooth is subjected to orthodontic tension, the bone surface will show an increase in:**
A. Howship's lacunae.
B. Multi-nucleated giant cells.
C. Osteoblast activity and osteoid deposition.
D. Osteoclast-mediated resorption.
E. TRAP-positive cells.
43. **(Tricky) Which structure facilitates the vascular communication between the periosteum and the Haversian canals?**
A. Canaliculi.
B. Volkmann’s canals.
C. Sharpey’s fibers.
D. Resting lines.
E. Reversal lines.
44. **(Tricky) During the aging process, the alveolar bone may show a decrease in "cellularity." Which of the following is an expected age-related change in the bone?**
A. Increase in the width of the periodontal ligament.
B. Increase in the number of osteoblasts.
C. Smoother socket walls.
D. Sclerosis of the bone and narrower marrow spaces.
E. Transformation of all lamellar bone into woven bone.
45. **(Tricky) The "Cribriform Plate" is perforated by numerous openings. What is the functional significance of these perforations?**
A. They allow for the attachment of the gingival epithelium.
B. They provide a route for nerves and blood vessels to pass between the bone and the PDL.
C. They decrease the weight of the skull.
D. They are remnants of the dental lamina.
E. They are sites of future tooth eruption.
46. **What is the approximate percentage of the organic matrix of bone that consists of non-collagenous proteins?**
A. 1%
B. 10%
C. 25%
D. 50%
E. 90%
47. **Bone resorption is a complex process. Which of the following is the first step in the degradation of bone by osteoclasts?**

A. Secretion of collagenase.
B. Creation of an acidic environment to dissolve the mineral phase.
C. Phagocytosis of osteocytes.
D. Formation of a reversal line.
E. Activation of alkaline phosphatase.

48. **The "Periosteum" is a double-layered connective tissue. The inner "osteogenic" layer is primarily responsible for:**
A. Sensory innervation of the bone.
B. Muscle attachment.
C. Providing the collagen for the PDL.
D. Providing a source of osteoblasts for bone growth and repair.
E. Secreting synovial fluid.
49. **Haversian canals are typically oriented in which direction relative to the long axis of the bone?**
A. Perpendicular.
B. Transverse.
C. Oblique.
D. Radially.
E. Parallel.
50. **Which of the following is a characteristic of "Spongy Bone" (Cancellous bone)?**
A. It contains Haversian systems throughout.
B. It is composed of a network of trabeculae with intervening marrow spaces.
C. It is found lining the tooth socket directly.
D. It has a higher mineral content than cortical bone.
E. It is always non-vascular.
51. **Regarding the "Alveolar Process," the maxilla differs from the mandible in that:**
A. The maxilla has thicker cortical plates.
B. The mandible has more cancellous bone in the molar region.
C. The maxilla generally has more porous cortical plates and more abundant cancellous bone.
D. The maxilla is formed by endochondral ossification.
E. The maxilla does not resorb after tooth loss.
52. **What is the primary role of "Osteocalcin" in the bone matrix?**
A. It acts as a lubricant.
B. It is a vitamin K-dependent protein that binds calcium and regulates mineralization.
C. It is the primary structural fiber.
D. It dissolves the mineral matrix.
E. It forms the reversal lines.
53. **The transition between the "alveolar bone proper" and the "supporting bone" is histologically marked by:**
A. The presence of the reversal line.
B. The appearance of trabecular bone or Haversian bone instead of bundle bone.
C. The presence of the mucogingival junction.
D. The termination of the dental lamina.
E. The presence of the epithelial attachment.
54. **"Osteopontin" is a non-collagenous protein involved in:**
A. Dentin formation only.
B. Cell adhesion and the anchoring of osteoclasts to the bone surface.
C. Increasing the transparency of enamel.
D. The formation of the secondary palate.
E. The inhibition of all mineralization.
55. **The term "Osteon" is synonymous with:**
A. Osteoblast.
B. Osteoclast.

C. Haversian system.
D. Howship’s lacuna.
E. Canaliculi.

56. **In the "Incisor region" of the mandible, the alveolar process is characterized by:**
A. Very thick cancellous bone between the plates.
B. Thin cortical plates with little to no intervening cancellous bone.
C. The presence of the maxillary sinus.
D. Absence of a cribriform plate.
E. The presence of secondary dentine.
57. **Which of the following describes the "clear zone" of an active osteoclast?**
A. The area containing the nuclei.
B. The region rich in mitochondria.
C. The area where the cell is attached to the bone, sealing the resorption site.
D. The site of collagen synthesis.
E. The part of the cell that secretes alkaline phosphatase.
58. **The "Alveolar Bone Proper" is embryologically derived from:**
A. The dental papilla.
B. The enamel organ.
C. The dental follicle (dental sac).
D. The vestibular lamina.
E. The successional lamina.
59. **Which of the following is true regarding the "Remodeling" of alveolar bone?**
A. It occurs only during childhood.
B. It involves only the formation of bone, never resorption.
C. It is a continuous process throughout life to maintain calcium homeostasis and structural integrity.
D. It is inhibited by orthodontic force.
E. It only occurs in the cortical plates.
60. **The "Basolateral Membrane" of the osteoclast is responsible for:**
A. Anchoring to the bone.
B. Secretion of acid.
C. Exocytosis of degraded bone products.
D. The formation of the ruffled border.
E. Mineralizing the osteoid.
61. **What is the significance of "Resting Lines" in a histological section of bone?**
A. They indicate a period of active resorption.
B. They show the location of nerves.
C. They represent a pause in bone deposition.
D. They are found only in the cementum.
E. They represent a fracture site.
62. **The mineral crystals of bone are typically located:**
A. Only within the osteocytes.
B. Within and between the collagen fibrils.
C. Only on the surface of the periosteum.
D. Within the marrow spaces.
E. In the cytoplasm of osteoclasts.
63. **In clinical practice, the "width" of the alveolar process is most relevant for:**
A. Determining the color of the teeth.
B. The placement of dental implants.
C. Calculating the rate of saliva flow.
D. Determining the taste sensation of the tongue.
E. Measuring the depth of the sulcus terminalis.
64. **Which enzyme is specifically used as a histological marker to identify osteoclasts?**

A. Alkaline phosphatase.
B. Amylase.
C. Tartrate-resistant acid phosphatase (TRAP).
D. DNA polymerase.
E. Lysozyme.

65. **The "Ruffled Border" of an osteoclast is:**
A. The site of new bone matrix secretion.
B. The highly folded plasma membrane where bone resorption occurs.
C. The part of the cell containing the Golgi apparatus.
D. Found only in resting osteocytes.
E. Absent during the resorption process.
66. **In the maxilla, the "Alveolar Bone Proper" is often very thin. On the lingual aspect of the maxillary molars, it usually fuses with:**
A. The nasal floor.
B. The zygomatic bone.
C. The cortical plate of the hard palate.
D. The soft palate.
E. The coronoid process.
67. **Which of the following factors can stimulate alveolar bone resorption?**
A. Increased mechanical load (within physiological limits).
B. Parathyroid hormone (PTH).
C. Calcitonin.
D. High intake of Vitamin C.
E. Estrogen therapy.
68. **The final "mature" state of bone that replaces woven bone is:**
A. Osteoid.
B. Cartilage.
C. Lamellar bone.
D. Bundle bone only.
E. Dentin.

Answer Key (31–68)

Question	Answer	Question	Answer	Question	Answer
31	D	44	D	57	C
32	B	45	B	58	C

33	D	46	B	59	C
34	B	47	B	60	C
35	D	48	D	61	C
36	E	49	E	62	B
37	C	50	B	63	B
38	A	51	C	64	C
39	A	52	B	65	B
40	A	53	B	66	C
41	C	54	B	67	B
42	C	55	C	68	C
43	B	56	B		

MCQ Analysis 😱

Q16: The term "Dehiscence" refers to a specific defect in the alveolar bone...

- **Correct Answer: C** (A V-shaped or cleft-like loss of cortical bone including the marginal edge)
- **Explanation:** In clinical periodontology, it is vital to distinguish between a *fenestration* (a window where the marginal bone remains intact) and a *dehiscence* (where the bone loss extends through the marginal crest). A dehiscence typically results in a recessed appearance of the bone over the root surface.

Q33: In the histology of bone, "interstitial lamellae" represent...

- **Correct Answer: D** (Remnants of older Haversian systems left behind during remodeling)
- **Explanation:** Bone is dynamic. When new osteons (Haversian systems) are formed, they cut through older ones. The fragmented remains of those previous circular layers that fill the gaps between new osteons are termed interstitial lamellae.

Q41: In a case of chronic periodontitis, the "Lamina Dura" may appear broken or absent...

- **Correct Answer: C** (The cribriform plate / alveolar bone proper)
- **Explanation:** This bridges the gap between radiology and histology. The "Lamina Dura" is purely a radiographic term. Its histological equivalent is the *alveolar bone proper* or *cribriform plate*. When it disappears on an X-ray, it signifies the loss of the dense bone lining the socket.

Q42: If a tooth is subjected to orthodontic tension, the bone surface will show an increase in...

- **Correct Answer: C** (Osteoblast activity and osteoid deposition)
- **Explanation:** Orthodontic movement relies on the "Pressure-Tension Theory." On the side where the PDL is stretched (tension), osteoblasts are recruited to lay down new bone (osteoid). Conversely, the "pressure" side triggers osteoclasts to resorb bone.

Q43: Which structure facilitates the vascular communication between the periosteum and the Haversian canals?

- **Correct Answer: B** (Volkmann’s canals)
- **Explanation:** While *Haversian canals* run longitudinally (parallel) to the long axis of the bone, *Volkmann’s canals* run transversely (perpendicular). They are the "supply lines" that bring blood vessels from the outer periosteum or inner endosteum into the Haversian systems.

Q53: The transition between the "alveolar bone proper" and the "supporting bone" is marked by...

- **Correct Answer: B** (The appearance of trabecular bone or Haversian bone instead of bundle bone)

- **Explanation:** The alveolar bone proper is characterized specifically by *bundle bone* (containing Sharpey’s fibers). Once you move deeper into the "supporting bone," the histology shifts to either organized lamellar/Haversian bone (in the cortical plates) or trabecular bone (in the cancellous area).

Q60: The "Basolateral Membrane" of the osteoclast is responsible for...

- **Correct Answer: C** (Exocytosis of degraded bone products)
- **Explanation:** An active osteoclast is polarized. The *ruffled border* faces the bone and handles resorption/secretion of acid. The *basolateral membrane* (the side facing away from the bone) is responsible for releasing the digested mineral and collagen fragments into the extracellular space/circulation.

(The End)